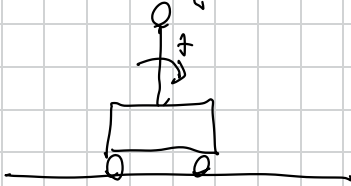


Simulation:

State is $x = \begin{bmatrix} p \\ \dot{p} \\ \theta \\ \dot{\theta} \end{bmatrix}$



$$M = 1.0 \text{ kg}$$

$$m = 0.1 \text{ kg}$$

$$l = 0.5 \text{ m}$$

$$g = 9.81 \text{ m/s}^2$$

$$\Delta t = 0.01 \text{ s}$$

Given: $(M+m)\ddot{p} + ml\dot{\theta} \cos \theta - ml\dot{\theta}^2 \sin \theta = F$

$$l\ddot{\theta} + \dot{p} \cos \theta - g \sin \theta = 0$$

↓

$$l\ddot{\theta} = g \sin \theta - \dot{p} \cos \theta$$

$$(M+m)\ddot{p} + mg \sin \theta \cos \theta - m\dot{p} \cos^2 \theta - ml\dot{\theta}^2 \sin \theta = F$$

$$(M+m)\ddot{p} - m\dot{p} \cos^2 \theta = F - mg \sin \theta \cos \theta + ml\dot{\theta}^2 \sin \theta$$

$$\ddot{p} (M+m) - m \cos^2 \theta = F - mg \sin \theta \cos \theta + ml\dot{\theta}^2 \sin \theta$$

$$\ddot{p} = \frac{F - mg \sin \theta \cos \theta + ml\dot{\theta}^2 \sin \theta}{M+m - m \cos^2 \theta}$$

$$M+m - m \cos^2 \theta$$

$$\ddot{\theta} = \frac{g}{l} \sin \theta - \frac{F - mg \sin \theta \cos \theta + ml\dot{\theta}^2 \sin \theta}{M+m - m \cos^2 \theta} \cdot \frac{\cos \theta}{l}$$